Chapter 34

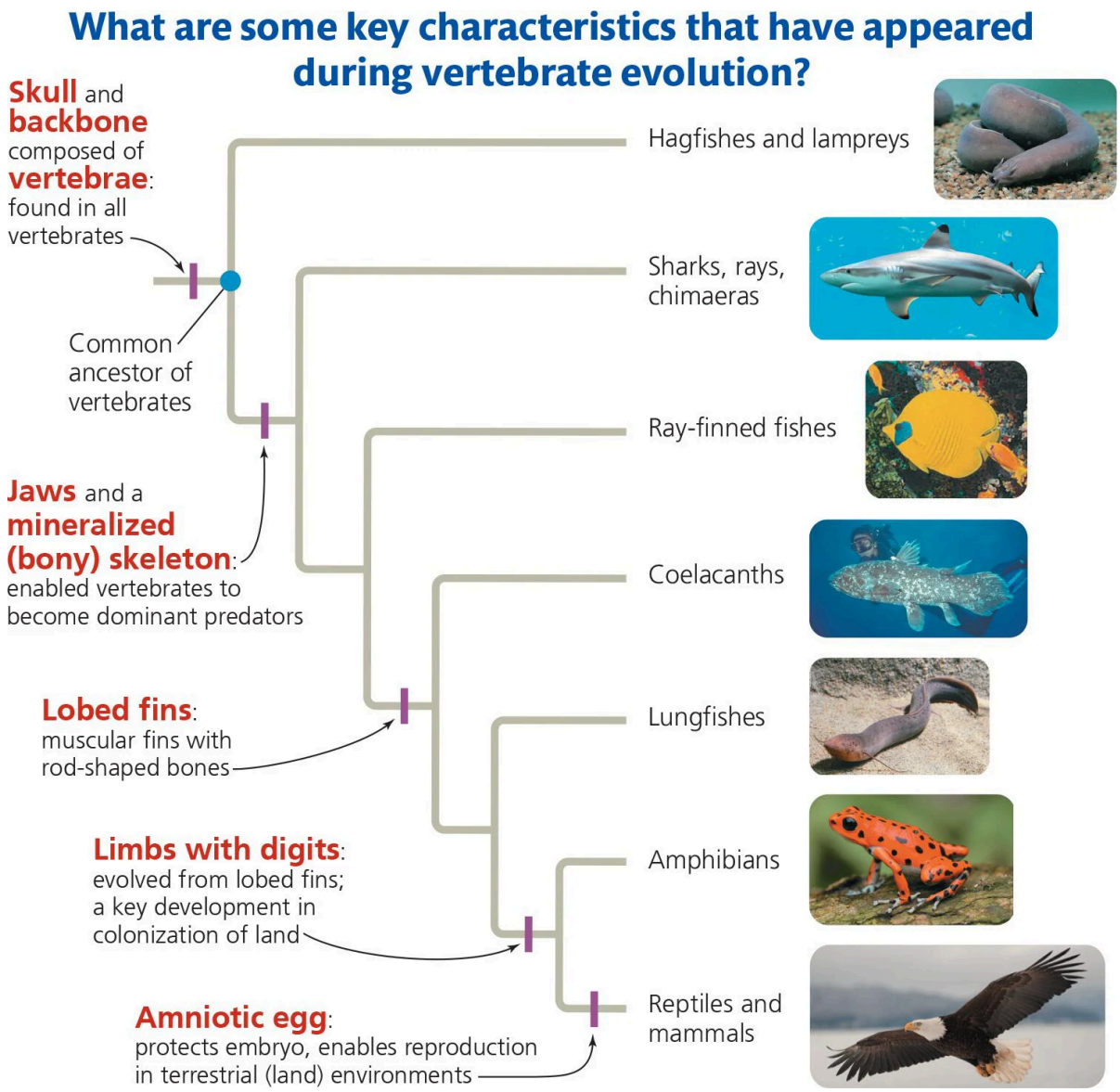
Campbell Biology, AP Edition, 12e

Chapter 34

The Origin and Evolution of Vertebrates

Figure 34.1

Small and slender, this inconspicuous species (*Myllokunmingia fengjiaoa*) lived in the ocean 530 million years ago, along with an immense variety of dangerous and well-defended invertebrate animals. Although lacking claws or armor, this ancient species was closely related to one of the most successful groups of animals on Earth: the vertebrates, animals with a backbone.



An evolutionary tree is given with key points labeled. Begin where skull and backbone composed of vertebrae are found in all vertebrates. After this is the common ancestor of all vertebrates. One branch goes to the following: jaws and a mineralized (bony) skeleton enabled vertebrates to become dominant predators. From here is a branch that leads to another branch. This branch splits and goes through the following: lobed fins - muscular fins with rod-shaped bones. This splits to 2 branches, one of which splits again, going through the following: limbs and digits evolved from lobed fins; a key development in colonization of land. After this is a branch that splits to amphibians and to the following key point: amniotic egg protects embryo; enables reproduction in terrestrial (land) environments. This leads to reptiles and mammals.

Key Concepts

- **34.1Chordates have a notochord and a dorsal, hollow nerve cord****

- **34.2Vertebrates are chordates that have a backbone****
- **34.3Gnathostomes are vertebrates that have jaws****
- **34.4Tetrapods are gnathostomes that have limbs****
- **34.5Amniotes are tetrapods that have a terrestrially adapted egg****
- **34.6Mammals are amniotes that have hair and produce milk****
- **34.7Humans are mammals that have a large brain and bipedal locomotion****

Natural selection within the phylum Chordata led to the evolution of a great variety of species whose bodies are supported by a backbone **(BIG IDEA 1)**. This adaptation has enabled vertebrates to compete successfully for resources on land, in the sea, and in the air **(BIG IDEA 4)**.

Study Tip

Make a table: Humans are mammals—and we are also amniotes, tetrapods, lobe-fins, osteichthyans, gnathostomes, vertebrates, and chordates (see Figure 34.2). As you read the chapter, list derived characters for these clades and describe whether these traits have been retained, modified, or lost in the human lineage.

Clade	Derived characters	Found in humans?
Chordates	Notochord	Modified (gelatinous disks between vertebrae)
	Post-anal tail	Lost

The table is as follows:

Clade	Derived characters	Found in humans?
Chordates	Notochord	Modified (gelatinous disks between vertebrae)
Chordates	Post-anal tail	Lost
blank	blank	blank

There are more than 60,000 species of **vertebrates**, animals with a backbone. This is a relatively small number compared to, say, the 1million insect species on Earth. But what vertebrates lack in number of species, they make up for in *disparity*, varying enormously in characteristics such as body mass. Vertebrates include the heaviest animals ever to walk on land, plant-eating dinosaurs that were as massive as 40,000 kg (heavier than 13 pickup trucks). The biggest animal ever to exist on Earth is also a vertebrate—the blue whale, which can exceed 100,000 kg. On the other end of the spectrum, the fish *Schindleria brevipinguis* is just 8.4 mm long and has a mass roughly 100billion times smaller than that of a blue whale.

In this chapter, we will track the origin and evolution of the vertebrate body plan, from a notochord to a head to a mineralized skeleton to limbs with digits (such as toes). We'll also explore the major groups of vertebrates (both living and extinct), as well as the evolutionary history of our own species—*Homo sapiens*.

Get Ready for This Chapter

In this chapter, you will learn about the origin, evolutionary history, and diversity of the vertebrates. Before beginning this chapter, you should be able to do the following:

- Describe the process of descent with modification (see Concept 22.3).

- Distinguish between shared derived and shared ancestral characteristics (see Concept 26.3, Figure 26.12).
- Identify monophyletic and paraphyletic groups (see Concept 26.3, Figure 26.10).

The following questions provide a quick check of your basic knowledge in these areas.

1. Which of the following statements is the most inclusive description of descent with modification?

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Similarity between two species results from genetic drift in small populations
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That's incorrect. Try again.

Incorrect. Review the following material: Evolution is a process of descent with modification: Characteristics present in an ancestral organism are altered in its descendants over time as they face different environmental conditions. One result of descent with modification is that related species can have characteristics that have an underlying similarity yet different functions. Descent with modification occurring in different evolutionary lineages may result in species in each of those lineages having characteristics that function similarly yet are not due to common descent as those species have independently adapted to similar environmental conditions.

```
Natural selection leads to the evolution of similar features in independent evoluti
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That's incorrect. Try again.

Incorrect. Review the following material: Evolution is a process of descent with modification: Characteristics present in an ancestral organism are altered in its descendants over time as they face different environmental conditions. One result of descent with modification is that related species can have characteristics that have an underlying similarity yet different functions. Descent with modification occurring in different evolutionary lineages may result in species in each of those lineages having characteristics that function similarly yet are not due to common descent as those species have independently adapted to similar environmental conditions.

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Similarity between two species results from convergent evolution.
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That's incorrect. Try again.

Incorrect. Review the following material: Evolution is a process of descent with modification: Characteristics present in an ancestral organism are altered in its descendants over time as they face different environmental conditions. One result of descent with modification is that related species can have characteristics that have an underlying similarity yet different functions. Descent with modification

occurring in different evolutionary lineages may result in species in each of those lineages having characteristics that function similarly yet are not due to common descent as those species have independently adapted to similar environmental conditions.

```
A trait present in an ancestral organism is modified by natural selection over time
```

That's correct.

Correct! If you had trouble with this question, review the following material: Evolution is a process of descent with modification: Characteristics present in an ancestral organism are altered in its descendants over time as they face different environmental conditions. One result of descent with modification is that related species can have characteristics that have an underlying similarity yet different functions. Descent with modification occurring in different evolutionary lineages may result in species in each of those lineages having characteristics that function similarly yet are not due to common descent as those species have independently adapted to similar environmental conditions.

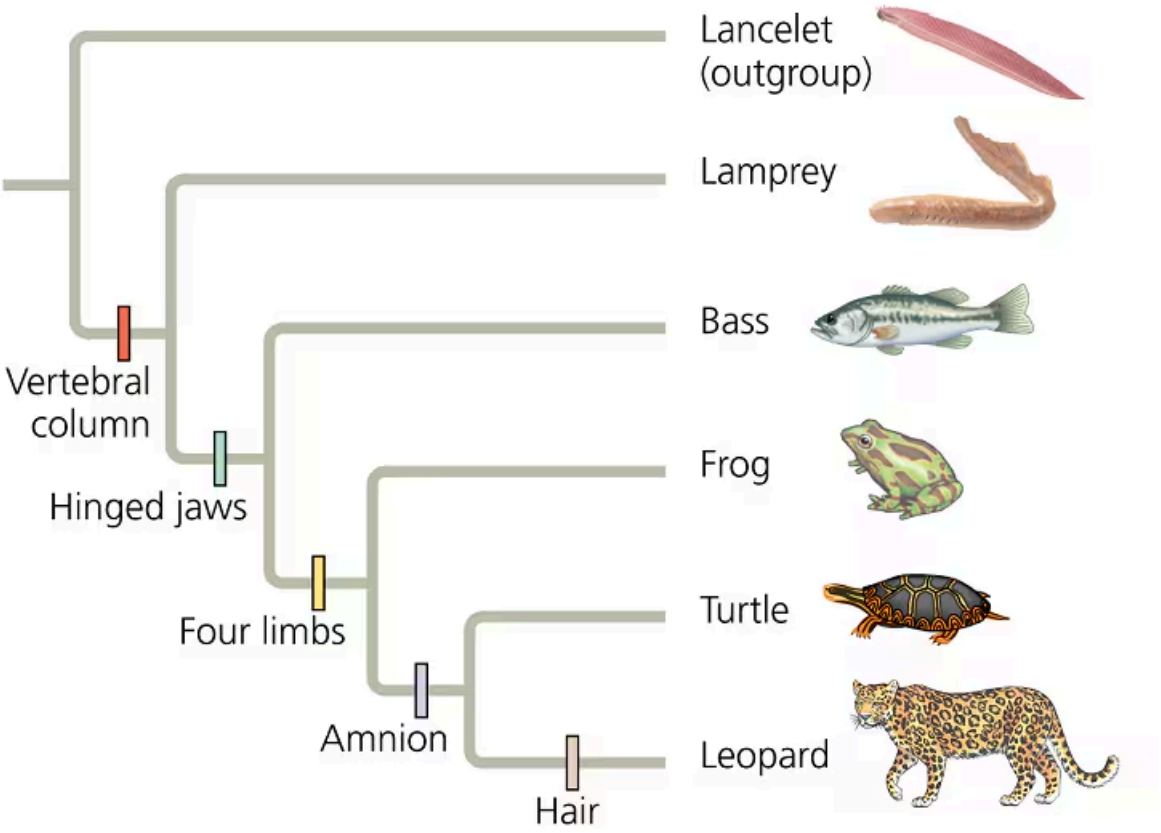

Diagram showing a series of horizontal forked branches. Reading left to right, the first branch point forks into a horizontal line leading to Lancelet (outgroup) and another horizontal line which has a vertical hatch mark labeled Vertebral column, and then branches again. This second branch point forks into a horizontal line leading to Lamprey and another horizontal line which has a vertical hatch mark labeled Hinged jaws,

and then branches again. This third branch point forks into a horizontal line leading to Bass and another horizontal line, which has a vertical hatch mark labeled Four limbs, and then branches again. This fourth branch point forks into a horizontal line leading to Frog and another horizontal line, which has a vertical hatch mark labeled Amnion, and then branches again. This fifth branch point forks into a horizontal line leading to Turtle and another horizontal line, which has a vertical hatch mark labeled Hair and leads to Leopard.

2. In the phylogenetic tree shown, which of the following are shared ancestral characters for members of the turtle-leopard group?

```
hinged jaws; amnion
```

That's incorrect. Try again.

Incorrect. Review the following material: A shared ancestral character is a character that is not unique to a particular group, having originated in an ancestor of that group. In contrast, a shared derived character is an evolutionary novelty that is unique to a particular group; in the phylogeny shown here, an amnion is a shared derived character for the turtle-leopard group.

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vertebral column; four limbs
```

That's correct.

Correct! If you had trouble with this question, review the following material: A shared ancestral character is a character that is not unique to a particular group, having originated in an ancestor of that group. For members of the turtle-leopard group, the phylogeny shows that a vertebral column, hinged jaws, and four limbs are all shared ancestral characters, having originated before the turtle-leopard group diverged from other lineages shown in the tree. In contrast, a shared derived character is an evolutionary novelty that is unique to a particular group; in the phylogeny shown here, an amnion is a shared derived character for the turtle-leopard group.

```
four limbs; hair
```

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vertebral column; amnion
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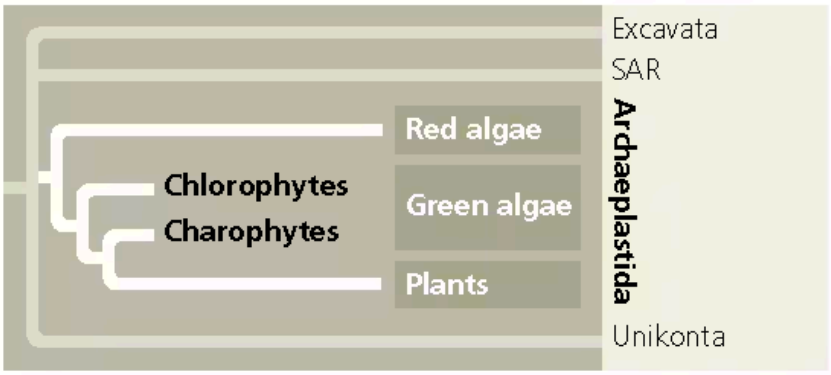

Diagram showing a series of white horizontal forked branches. Reading left to right, the first branch point forks into a horizontal line leading to Red algae and another horizontal line, which branches again. This second branch point forks into a horizontal line leading to Chlorophytes and another horizontal line, which branches again. This third branch point forks into a horizontal line leading to Charophytes and another horizontal line leading to Plants. The Chlorophyte and Charophyte branches together are labeled Green algae. At the far right of the diagram a vertical label Archaeplastida extends from the Red algae down through the Plants.

3. Based on the highlighted (white) branches in the phylogenetic tree shown, which of the following labels refers to a paraphyletic group?

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green algae
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That's correct.

Correct! If you had trouble with this question, review the following material: A **monophyletic** group consists of an ancestral species and all of its descendants. Examples of monophyletic groups in this phylogenetic tree include red algae, plants, charophytes, chlorophytes, and the supergroup Archaeplastida. In contrast, a **paraphyletic** group consists of an ancestral species and some, but not all, of its descendants. In this tree, green algae are a paraphyletic group, because they consist of chlorophytes, charophytes, and their common ancestor, but not plants, which are also descended from the same common ancestor.

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red algae
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That's incorrect. Try again.

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```
plants
```

That's incorrect. Try again.

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Archaeplastida
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That's incorrect. Try again.

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Learning Objectives

After studying this chapter, you should be able to do the following:

- 34.1Identify key derived characters of chordates.

- 34.2Describe the evolutionary history of vertebrates.
- 34.3Describe gnathostomes and explain the evolutionary advantages of having jaws.
- 34.4Explain how the first tetrapods may have arisen.
- 34.5Identify key derived characters of amniotes and give examples of amniote groups.
- 34.6Differentiate between monotremes, marsupials, and eutherian mammals.
- 34.7 Identify key hominin lineages and describe their characteristics.